

Genome-wide analysis of lectin receptor-like kinase family from *Arabidopsis* and rice

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Abstract Lectin receptor-like kinases (LecRLKs) are class of membrane proteins found in higher plants that are involved in diverse functions ranging from plant growth and development to stress tolerance. The basic structure of LecRLK protein comprises of a lectin and a kinase domain, which are interconnected by transmembrane region. Here we have identified LecRLKs from *Arabidopsis* and rice and studied these proteins on the basis of their expression profile and phylogenies. We were able to identify 32 G-type, 42 L-type and 1 C-type LecRLKs from *Arabidopsis* and 72 L-type, 100 G-type and 1 C-type LecRLKs from rice on the basis of their annotation and presence of lectin as well kinase domains. The whole family is rather intron-less. We have sub-grouped the gene family on the basis of their phylogram. Although on the basis of sequence the members of each group are closely associated but their functions vary to a great extent. The interacting partners and coexpression data of the genes revealed the importance of gene family in physiology and stress related responses. An in-depth analysis on gene-expression suggested clear demarcation in roles assigned to each gene. To gain additional knowledge about the LecRLK gene family, we searched for previously unreported motifs and checked their importance structurally on the basis of homology modelling. The analysis revealed that the gene family has important roles in diverse functions in plants, both in the developmental stages and in stress

conditions. This study thus opens the possibility to explore the roles that LecRLKs might play in life of a plant.

Keywords *Arabidopsis* · Genome-wide analysis · Lectin receptor-like kinase · Motif identification · Phylogenetic analysis · Rice · Stress · Transcriptomics

Introduction

Darwin suggested survival of the fittest. Plants are in a constant state of war against an array of pathogens and adverse environmental factors. To survive these, plants are geared up with their knights in action, in the form of receptor-like kinases (RLKs). These RLKs gather information about friends and foes, both biotic as well abiotic, and translate the perceived information into a biochemical response to aid survival of the plant. The RLKs are endowed with an N terminus extracellular domain that collects information by sensing presence of a very specific ligand and passes that information to its C terminus cytoplasmic domain which is a kinase, the effector portion of the molecule. The two regions are in contact with each other via a transmembrane region, which hooks the molecule to the cellular membrane system. The highly variable N terminus determines the specificity of the response of the RLK whereas the kinase domain shares high degree of conservation amongst various species. Therefore, the classification of RLKs has been done on the basis of their extracellular domain. Of the 15 classes of RLKs that have been described (Shiu and Bleecker 2001), here we are focussing only on the Lectin receptor-like kinase (LecRLK) family. This family was first described by Herve et al. 1996, in *Arabidopsis thaliana*. No homologues of Lectin kinase have been found or reported in yeast and humans

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(Wan et al. 2008; Navarro-Gochicoa et al. 2003). Other plants in which this protein has been studied include *Lombardy poplar* (Nishiguchi et al. 2002), *Oryza sativa* (Chen et al. 2006), *Nicotiana benthamiana* (Kanzaki et al. 2008), *Nicotiana tabacum* (Sasabe et al. 2007), *Nicotiana attenuate* (Bonaventure 2011; Gilardoni et al. 2011) and amongst legumes in *Medicago truncatula* (Navarro-Gochicoa et al. 2003) and *Pisum sativum* (Joshi et al. 2010). A study carried out by Hervé et al. 1999 suggests that a higher homology exists between LecRLKs of dicots as compared to monocots.

All LecRLKs possess two definitive domains which define this class of protein, a C terminal kinase domain, which as discussed above, remains fairly conserved amongst different genera and an N terminal lectin domain. Depending on the variability of N terminal lectin domain, this class has been subgrouped in 3 sub-classes, G, L and C type (Shiu and Bleecker 2001). G type LecRLKs possess α -D mannose specific plant lectins. It is also accompanied in most of the protein (not necessarily all) by both or either of an EGF and/or a PAN motif. EGF motif is cysteine rich (Shiu and Bleecker 2001) and probably takes part in the formation of disulfide bonds. The PAN motif is believed to be involved in protein-protein and protein carbohydrate interactions (Tordai et al. 1999; Loris 2002; Naithani et al. 2007). As the name suggests the lectin domain of L type LecRLKs resembles soluble lectin proteins found in leguminous plants (Hervé et al. 1999). The third class of LecRLKs is C type lectin kinases. The C type lectin kinases in plants are thought to be homologues of calcium-dependent lectin motifs which is a large group of mammalian proteins known to be involved in innate immune responses and in pathogen recognition (Cambi et al. 2005). Though the C type LecRLKs are present in large number in mammalian system, both rice and arabidopsis have only one gene encoding for C type LecRLK protein (Bouwmeester and Govers 2009).

The role of LecRLK in a plant seems to be varied and diverse. It is believed to be involved in plant development and wounding (Garcia-Hernandez et al. 2002; Riou et al. 2002; Nishiguchi et al. 2002), in fibre development in cotton plant (Zuo et al. 2004), in process of pollen development (Wan et al. 2008), in plant symbiotic associations (Navarro-Gochicoa et al. 2003; Herve et al. 1996; Hirsch 1999), in induction of senescence associated and other plant defence associated genes (Ohtake et al. 2000), in providing hypersensitivity response (HR) upon attack of an elicitor (Kanzaki et al. 2008), in conferring resistance to rice against fungal pathogen (Chen et al. 2006), bacterial pathogens (Singh et al. 2012), herbivory by insects (Gilardoni et al. 2011), perception of insect feeding (Bonaventure 2011), in providing salinity stress tolerance (Deng et al. 2009; Joshi et al. 2010) and in regulation of stomatal immunity in Arabidopsis (Desclos-Theveniau et al. 2012). The availability of complete

genomes of the two most widely studied plants, Rice (Rice genome Annotation Project database) (<http://rice.plantbiology.msu.edu/>) (Goff et al. 2002; Yu et al. 2002) and Arabidopsis (<http://www.arabidopsis.org>, The Arabidopsis Genome Initiative 2000), has opened up wide avenues for comparative analysis of a particular protein in these two plants and probably apply the knowledge to other proteins of interest as well.

The objective of the current study was to survey the *LecRLK* genes in fully sequenced model plants: rice and arabidopsis. Using bioinformatics approach, the information derived from publically available plant databases was used to analyze phylogenetic relationship between the genes of the LecRLK family. The sequence data was employed to search for novel motifs in lectin domain which, in future, could be used for functional studies of this gene family. Also, the gene expression data, using publically available expression databases was used to analyze the probable role of *LecRLK* gene family.

Materials and methods

Exploration of database, protein properties and domain organization

To fish out LecRLK genes, initial search was carried out using At4G04960 as query locus for L type LecRLKs in The Arabidopsis Information Resource (TAIR) databases (<http://www.arabidopsis.org/>) (Poole 2007). The results obtained with $e\text{-value} \leq 0.1$ were chosen as query locus ids for next rounds of BLAST. The search was carried out till all the locus ids found were used as query locus. For arabidopsis G type LecRLK, At1G61610 was used as initial query locus as it was obtained as one of the candidates while carrying out search for L type LecRLK with $e\text{-value} < 1$.

To search for rice LecRLK genes, nucleotide sequence of At4G04960 was used as query for BLAST in Rice genome annotation project (<http://rice.plantbiology.msu.edu/>) (Ouyang et al. 2007) and resulting gene matches were used as query sequences for further search as described above. E-value cut off of 0.1 was used for all searches. These candidates were sorted into L-type, G-type and C-type LecRLK from both the plant groups on the basis of identity of lectin domain. LecRLKs with legume lectin family and concanavalin A-like lectin/glucanase identity were classified as L type LecRLKs while those with bulb lectin domains or S domain locus and α -D mannose binding affinity were classified as G type LecRLKs. All the sequences were manually checked for the presence lectin as well as the kinase domains using <http://prosite.expasy.org/>. The sequences which lacked either of the domains were excluded from the search. Dataset and details of Arabidopsis and rice genes was retrieved from TAIR (<http://www.arabidopsis.org/>)

and Rice genome annotation project database (<http://rice.plantbiology.msu.edu/>). Co-expressed genes for Arabidopsis *LecRLK* genes were predicted using ATTED-II (<http://atted.jp/>). Prediction of interacting partners of Arabidopsis *LecRLK* gene family was made using Arabidopsis thaliana Protein Interactome Database (<http://atpid.biosino.org/>). The localization data for Arabidopsis genes was retrieved from TAIR database and cross checked with experimentally proven results of more up to date SUBA database (<http://suba.plantenergy.uwa.edu.au/>) while that of rice was predicted using CELLO v.2.5: sub-CELLular LOcalization predictor (<http://cello.life.nctu.edu.tw/>) (Yu et al. 2004, 2006) and cross checked using pSORT prediction software (<http://wolfsort.org/>) (Horton et al. 2007). Apart from this, chloroplast localization was also confirmed by ChloroP (<http://www.cbs.dtu.dk/services/ChloroP/>) (Emanuelsson, 1999).

Genome organization

To visualize *LecRLK* loci on Arabidopsis and rice chromosomes, chromosomal maps were constructed. For Arabidopsis *LecRLK* genes, chromosomal map tool available on TAIR database (<http://arabidopsis.org/jsp/ChromosomeMap/tool.jsp>) was used while to construct rice chromosomal map, Oryzabase- Integrated Science Database was used (<http://viewer.shigen.info/oryzavw/maptool/MapTool.do>). We also checked the distance between genes with closely lying loci using the chromosomal locations given by TAIR databases (<http://www.arabidopsis.org/>) for Arabidopsis and Rice genome annotation project (<http://rice.plantbiology.msu.edu/>) for rice genes.

Protein alignment and phylogenetic analysis

Protein sequences of 74 Arabidopsis and 172 rice *LecRLKs* were downloaded from TAIR and Rice genome annotation project databases respectively and were aligned using ClustalW using default parameters. Phylogenetic analysis was conducted using MEGA version 5 (Tamura et al. 2011). To build phylogenies, neighbour joining method (Saitou and Nei 1987) was employed and bootstrap analysis was conducted using 1000 replicates. The phylogram was rooted using distantly related RLK sequences of Arabidopsis RLKs, WAK1 (At1g21250), PERK1 (At3g24550), ARK1 (At1g65790) and CES101 (At3g16030) and a lectin protein, Lectin 3.1 (At3g15356).

Generation and identification of motifs

To identify conserved motifs within lectin domain of L type and G type *LecRLKs*, MEME Suite-GLAM2 version 4.8.0 (Bailey et al. 2009) was employed (<http://meme.nbcr.net/meme/intro.html>) using default settings.

Homology modelling for lectin domains

Homology model of lectin domain of L type *LecRLK* was created using ESyPred3D Web Server 1.0 (<http://www.fundp.ac.be/sciences/biologie/urbm/bioinfo/esypred/>). Since the same software could not find a homologue for lectin domain of G type *LecRLK*, ab initio method provided by i.Tasser online protein structure and function prediction software (<http://zhanglab.ccmb.med.umich.edu/I-TASSER/>) was used. To visualize and edit the PDB models provided by the two softwares, molecular graphics visualization program PyMol was used.

In silico gene expression analysis

For analysis of expression pattern of *LecRLK* gene family in various development stages, in different organs and in various stress conditions, expression values were obtained from Affymetrix array databases using Genevestigator Response Viewer (<https://www.genevestigator.com/gv/>) (Hruz et al. 2008). For Arabidopsis, ATH1-22 k array was chosen and microarray data of only wild type Arabidopsis was considered for further analysis. In case of rice, microarray results of OS_51 K: Rice Genome 51 K array were analysed. For construction of heat map, hierarchical clustering tool was chosen. This tool identifies a group of genes that are co-regulated across given conditions. For distance measurement in hierarchical clustering Pearson correlation coefficient was selected.

Results

Exploration of database, protein properties and domain organization

Post screening, sequences of 42 L- type *LecRLKs* and 32 G-type *LecRLKs* from Arabidopsis (Tables 1, 2) and 72 L- type *LecRLKs* and 100 G-type *LecRLKs* from rice (Tables 3, 4) were retrieved from TAIR and Rice genome annotation project databases respectively. Only one C-type *LecRLK* has been reported in both Arabidopsis and rice genome databases. During screening and analysis of the sequences, few discrepancies were observed in Rice genome annotation project database. LOC_Os01g40540.1 and LOC_Os02g32680.1 have been annotated as lectin receptor-type protein kinase, but the proteins neither have lectin nor kinase domain. Similarly LOC_Os09g17010 and LOC_Os01g65060 lack lectin domain while LOC_Os01g12400 has PAN domain but no lectin domain signature but all proteins have been named as lectin protein kinase. LOC_Os11g03860 has PAN and S locus lectin domain but no kinase domain but it has been named as Ser/Thr kinase.

Table 1 *Arabidopsis thaliana* L type lectin receptor-like kinase

Gene locus	SF	CDS	Mol. Mass (kDa)	pI	Co-expressed genes	Interacting partners	Localization
At4G04960	0	2331	76.69	6.6	LRR kinase (At3g28450); LECRK4.2 (At5g01550)	PPR superfamily protein (AT5G04000)	EM
At4G28350	0	1950	72.07	6.2	Zinc finger family protein (At3g13430), peptidoglycan-binding LysM domain-containing protein (At2g23770); LRR kinase (At1g74360); leucine zipper protein 18 (At1g70920)	Not Reported	PM ^a
At4G29050	0	2062	74.74	7.2	Protein kinase family protein (At5g38210; At5g39020). ATCNGC13 (At4g01010), disease resistance protein (At1g58410, At5g02630)	LecRLK (AT4G02420), TATA-binding-protein associated factor TAF5 (AT5G25150)	EM
At5G60280	0	1974	72.39	8.4	Transmembrane receptor (At5g45000), Nudix hydrolase (At2g04430), disease resistance protein (At3g14470), LecRLK (At3g45330)	Pentatricopeptide repeat superfamily protein (AT1G11710)	EM
At4G02410	0	2265	75.92	7.9	AT-HSFA4A (At4g18880), LRR kinase (At2g31880), protein kinase family (At5g38210)	LecRLK (AT5G59260)	EM
At4G02420	0	2010	75.4	8.7	Protein kinase family protein (At1g69730; At1g29720), serine carboxypeptidase-like 46 (At2g33530), unknown protein (At2g24160)	LecRLK (AT3G55550, AT3G53810, AT3G45440), LRR-RLK (AT5G67200), TAF5 (AT5G25150)	EM
At2G37710	0	2287	75.54	8.0	Protein phosphatase 2C (At1g34750), MEK1 (At4g26070), kinase (At4g08850)	Protein phosphatase (AT5G19280), G-Box Factor(AT5G65430), LecRLK (AT5G60320, AT5G65600)	PM
At2G29220	2	1884	68.79	8.4	UDP-glucosyl transferase (At2g26480), heavy-metal-associated domain-containing protein (At3g04900), unknown protein (At3g09280)	Ankyrin repeat family protein (AT1G05640), PPR superfamily protein (AT2G15630, AT2G35030), LRR-RLK (AT3G02880), WPP Domain-interacting protein (AT1G68910)	EM
At2G29250	0	1872	68.22	8.3	UDP-glucosyl transferase family protein (At3g02100), unknown protein (At3g43820, At1g36756)	PPR superfamily protein (AT3G11460)	EM
At3G59700	0	2143	74.44	6.7	LRR kinase (At1g74360), LecRLK (At3g16030), WRKY6 (At1g62300), Mildew resistance locus (At1g61560)	LecRLK (AT3G59750)	EM
At1G70110	0	2001	74.58	7.0	Transposable element gene (At1g53810; At2g05660), unknown protein (At2g30380; At3g56280; At2g44000)	LecRLK (AT1G15530, AT1G70130), PPR repeat-containing protein (AT1G64100)	EM
At3G45330	4	2049	74.91	8.8	CHP-rich zinc finger protein (At2g42060), LecRLK (At5g60280), Receptor like protein (At1g17240)	LecRLK (AT5G01540, AT3G45440)	EM
		2460					PM
		2490					EM
		2498					EM
At3G45390	0	1815	66.65	8.6	Aspartyl protease family protein (At3g51360), ATGLR2.1 (At5g27100), DC1 domain containing protein (At3g59120)	Ankyrin repeat family protein (AT4G14400), LecRLK (AT5G60320), LRR-RLK (AT5G62710)	EM
At3G45440	2	2010	73.5	8.0	Transposable element gene (At1g41810; At2g05200), kinase family protein (At3g24540)	LecRLK (AT3G45330, AT4G29050, AT4G02420), TBP associated factor TAF5 (AT5G25150)	EM
		2028			Transposable element gene (At3g31920), transport protein sec61 (At1g78720), zinc finger family protein (At5g37230)	LecRLK (AT3G45390, AT2G37710, AT5G65600)	Pd
At3G55550	2	2069	76.18	6.0	DC1 domain-containing protein (At2g02700), transcription factor/zinc ion binding (At5g60470), Purple acid phosphatase 24 (At4g24890), transposable element gene (At3g29220)	LecRLK (AT3G53810, AT4G02420)	PM ^a
	0	1851	68.67	6.6	CYP94B2 (At3g01900), DC1 domain-containing protein (At1g55440), transferase family protein (At3g26040), unknown protein (At3g10830, At1g07860)	Ankyrin repeat containing protein (AT1G03670), LecRLK (AT5G60300, AT3G45430)	EM

Table 1 continued

Gene locus	SF	CDS	Mol. Mass (kDa)	pI	Co-expressed genes	Interacting partners	Localization
AT2G43700	0	3207	73.91	7.5	CHP-rich zinc finger protein (At2g37810), zinc finger protein (At5g63740), WRKY71 (At1g29860)	Not reported	EM
AT3G45410	0	2115	73.91	7.7	Zinc finger family protein (At5g22890), protein kinase family protein (At4g05200), ZIP2 (At5g59520)	Ankyrin repeat containing protein (AT3G01750), LecRLK (AT2G43690)	PM ^a
AT3G45430	0	2025	74.38	9.1	LecRLK (At5g10530), MYB domain protein (At40330), unkn own protein (At3g53170)	LecRLK (AT5G60310, AT5G60300), LRR-RLK (AT5G10290), PPR superfamily protein (AT2G02750)	EM
AT5G60270	0	2299	73.95	8.8	WAKL2 (At1g16130), Gamma-glutamyl transpeptidase I (At4g39640), unknown protein (At5g05190)	LecRLK (AT5G60280, AT5G59270), LRR-RLK (AT5G45840), Class I polyA-binding protein (AT1G71770)	PM ^a
AT3G59740	0	2072	73.54	6.9	Not reported	MPPR domain protein (AT3G16890, AT3G48810); LRR-RLK (AT5G20690)	EM
AT3G59730	0	1572	59.31	8.7	MD-2-related lipid recognition domain-containing protein (At5g23480), CYP705A5 (At5g47900), CYP705A25 (At1g50560)	Ubiquitin ligase (AT1G04510)	EM
AT3G08870	0	2227	77.6	6.9	RLK (At5g60900), HAE (At4g28490), ankyrin (At5g54610), ATPase family protein (At3g28540), protein kinase (At5g39030), LRR kinase (At1g66830), unknown protein (At1g76960)	LecRLK (AT5G01560, AT5G01540), PPR superfamily protein (AT2G19280)	EM
AT3G45420	0	2004	74.1	6.8922	Zinc finger family protein (At2g17975), D111/G-patch domain-containing protein (At2g42330), unknown protein (At3g04160; At1g14090)	Not reported	Mt ^a
AT5G59260	0	2025	74.48	8.5615	DC1 domain-containing protein (At2g37820), LRR kinase (At3g47110), LecRLK (At5g59270), pectinesterase family protein (At2g19150), unknown protein (At4g31875)	AT4G02410:LecRLK	EM
AT2G43690	0	1995	74.4	5.7	ATP binding/RNA polymerase II transcription factor (At3g52270), PIPK (At3g56960), regulator of chromosome condensation family protein (At3g23270), embryo sac development arrest 18 (At2g34920)	AT3G01750:Ankyrin repeat family protein; AT3G45410: RLK	NR
AT5G01550	0	2067	75.98	7.0	LecRLK (At5g01560; At4g04960), inflorescence deficient in abscission gene (At1g68765), unknown protein (At1g51920)	AT4G16390:SVR7; AT3G18020: PPR superfamily protein; AT4G22730: LRR-RLK	EM
AT5G01540	0	2494	75.87	9.5	Cysteine rich RLK (At4g23190), UDP-glucosyl transferase (At3g21780), protein kinase family protein (At4g23220; At4g23200)	LecRLK (AT3G08870, AT5G01560, AT3G45330)	PM
AT5G01560	0	2393	76.73	7.3	Short hypocotyl under blue gene (At4g25350), PP2C (At1g07160), MATE efflux family protein (At1g66780), VQ motif containing protein (At1g80450), LecRLK (At5g01550), Histone acetyl transferase (At1g55970), invertase/pectin methyltransferase inhibitor family protein (At3g47380), unknown protein (At5g48530)	LecRLK (AT3G08870, AT5G01540, AT5G42120, AT4G37170); PPR superfamily protein AT2G45340; LRR-RLK: AT5G27110	EM
AT1G15530	0	1971	73.15	4.9	C2 domain-containing protein (At2g21040), unknown proteins (At1g15830; At3g23850; At2g37830)	LecRLK(AT1G70130, AT1G70110); Ankyrin repeat family protein (AT3G04140); Mitochondrial editing factor (AT1G62260)	PM ^a , Vc
AT3G59750	0	1881	70.37	6.3	Not reported	RLK(AT3G59700, AT1G28340); PPR superfamily protein (AT2G17525)	EM
AT5G03140	0	2355	77.12	6.3	Zinc finger family protein (At5g63700), IQD28 (At1g14380), unknown protein (At5g58930)	LecRLK(AT5G5830); RLK (AT3G28890); PPR repeat-containing protein(AT5G46580)	EM
AT1G70130	0	1971	72.45	8.1	CYP715A1 (At5g52400), zinc finger family protein (At3g53600), LRR disease resistance protein (At2g26380), AP2 domain-containing transcription factor (At1g22810), MAPKKK (At5g55090), LRR kinase (At2g24130), transcription factor (At3g23220), WAKL5 (At1g16160), S-locus protein kinase (At1g61460), unknown protein (At5g03270)	LecRLK(AT1G15530, AT1G70110); Ankyrin repeat family protein(AT3G04140); Mitochondrial editing factor (AT1G62260)	EM

Table 1 continued

Gene locus	SF	CDS	Mol. Mass (kDa)	pI	Co-expressed genes	Interacting partners	Localization
AT5G55830	0	2189	75.84	6.5	NLI interacting factor (NIF) family protein (At5g45700), unknown protein (At5g01910); At5g15510	LecRLK(AT5G03140); Slow growth(AT2G22410)	EM
AT3G53380	0	2355	77.67	7.4	Structural constituent of ribosome (At3g20260), AGO4 (At2g27040), VPS 2.2 (At5g44560), Protein kinase (At5g11020)	Not Reported	PM ^a
AT5G42120	0	2076	76.56	5.9	SPR/RYanodine receptor (SPRY) domain-containing protein (At1g51450), F-box family protein-related (At1g49790), unknown protein (At2g15110)	LecRLK(AT5G01560);TPR-like superfamily protein (AT5G27110); LRR-RLK(AT2G45340); PPR superfamily (AT4G37170)	EM
AT5G10530	0	1956	72	7.4	LecRLK (At3g45430), Disproportionating enzyme (At5g64860), Chlororespiratory reduction 4 (At2g45350), Beta glucosidase 43 (At3g18070)	Not Reported	EM
AT5G06740	0	1959	72.87	7.3	WRKY72 (At5g15130), CHP-rich zinc finger protein (At2g42060), GLIP3 (At1g53990), unknown protein (At2g44010; At2g28570)	Ankyrin repeat family protein (AT4G03500); PPR superfamily protein (AT1G10910)	EM
AT5G65600	0	2028	75.29	5.9	WAK (At1g79680), pectinacetyltransferase family protein (At3g09410), U-box domain-containing protein (At5g37490), MATE efflux protein (At4g23030), unknown protein (At1g28190; At1g57650)	RLK(AT2G37710, AT5G60320)	EM
AT5G59270	0	2007	74.4	7.6	Lec RLK (At5g59260), protein kinase family (At1g66930), idesease resistance protein (CC-NBS class) (At3g46730)	LecRLK(AT5G60270); Class I polyA-binding protein (AT1G17770)	EM

Abbreviations: (For localized in): EM endomembrane system, Mt mitochondria, PM plasma membranem, Pd plasmodesma, Vc Vacuole, SF spliced form, CDS coding sequence

^a Refers to experimentally observed localization of protein that was reported by SUBA database

There are other proteins like Os04g07200 and Os11g10290 in which complete kinase domain is absent but their sequences has a protein kinase signature while Os01g73390 has a truncated kinase domain. We also found few members of LecRLK family which had some unusual characters. As an example, Os11g03820 has a plant organelle RNA recognition domain at its N terminal, which is not found in any other member of this family. Lectin domain of Os11g10260 starts from amino acid 431 and the initial 431 amino acids bear resemblance with retrotransposon protein ABA92006. In case of Arabidopsis, we found 2 LecRLK proteins (At4G28350 and At1G67520) which lacked transmembrane region.

The average molecular weight of rice G type LecRLK was 88.5 kDa while that of Arabidopsis G type was 94.7 kDa, which was higher than the L type proteins of rice and Arabidopsis, which were 75.8 and 74 kDa, respectively. This could be due to the presence of an additional PAN and/or EGF domains in G type LecRLKs. The range of pI of rice L and G type LecRLKs were 5.3–10.2 and 5.1–10.1 (Tables 3, 4) which was wider than that of Arabidopsis L type and G type LecRLKs which had a range of 4.8–9 and 5–8.4 respectively (Tables 1, 2). The localization of LecRLK family proteins was found to be mainly in endomembrane system for Arabidopsis, as reported by TAIR and SUBA databases (The localization of gene confirmed by SUBA database has been marked with an asterisk in the table). A few of the genes were found to be localized in chloroplast still fewer were predicted to be expressed in plasmodesma (At5G60320, At4G21380), vacuole (At1G15530, At4G21380), cytoplasm (At1G65800), mitochondria (At3G45420) and in pollen pistil interaction (At1G65790) (Tables 1, 2). Rice LecRLKs were mainly found to be localized in plasma membrane and extracellular localization by CELLO v.2.5: subCELLular Localization predictor (<http://cello.life.nctu.edu.tw/>) (Yu et al. 2004, Yu et al. 2006). Following similar pattern as that of Arabidopsis, very few genes were found to have mitochondrial (Os11g10260, Os05g34390), nuclear (Os09g07200, Os04g23700), vacuolar (Os2g48210) and cytoplasmic localization (Os6g14260) (Tables 3, 4). The results obtained from CELLO were cross checked using pSORT prediction software (<http://wolfsort.org/>) (Horton et al. 2007). A strikingly unique feature was observed for rice L type LecRLK wherein 25 out of 72 proteins were found to be localized in chloroplast. Chloroplast localization was also confirmed by ChloroP (<http://www.cbs.dtu.dk/services/ChloroP/>) (Emanuelsson et al. 1999). It was observed that there is a variation in predictions of subcellular localization of some proteins by different softwares. Upon analysis of LecRLK family of arabidopsis and rice for the presence of alternative spliced forms, we found that out of a total of 246 genes, representing this family in two genomes only 13 genes {1 in

Table 2 *Arabidopsis thaliana* G type lectin receptor-like kinase

Gene locus	SF	CDS	Mol. Mass (kDa)	pI	Co-expressed genes	Interacting partners	Localization
AT1G61550	0	2982	89.3	7.58	Peroxidase (At5g06730), protein kinase (At4g11480), NAC domain containing protein (At3g04420), S-locus protein kinase (At1g61490), unknown protein (At5g66670)	AT4G22200:Potassium transport,AT1G61490,AT1G61420: lectin protein kinase, AT1G30290:unknown protein	EM
AT1G61420	0	2681	90.4	7.28	Armadillo/beta-catenin repeat family protein (At5g67340), Receptor like protein 35 (At3g11080), S-locus protein kinase (At1g61490), protein kinase (At4g04490)	Not reported	EM
AT1G61480	0	2605	90.7	6.91	Disease resistance protein (TIR-NBS-LRR class) (At2g17060, At2g17050), Protein kinase (At5g49780)	AT1G61500: lectin protein kinase; AT1G11290: Chlororespiratory Reduction22	EM
AT1G61490	0	2415	89.9	7.44	LecRLK (At1g61420; At1g61550), UDP-glucosyl transferase family protein (At1g05670)	AT1G61550, AT1G61420, AT1G61500: lectin protein kinase; AT4G35130: TPR-like superfamily protein	EM
AT1G61500	0	2415	90.3	6.46	Subtilase family protein (At4g10550), beta-galactosidase (At4g29200), hAT dimerisation domain-containing protein (At3g13030)	AT1G61480, AT1G61490: lectin protein kinase; AT4G35130:Tetratricopeptide repeat (TPR)-like superfamily protein	EM, Ec ^a
AT1G61440	0	2379	89.0	7.86	UDP-glucosyl transferase (At2g23260), unknown protein (At1g33820; At2g20170)	AT4G30700: PPR superfamily protein; AT5G40170: receptor like protein 54	EM
AT1G61430	0	2909	90.2	8.45	Cytochrome P450 (At2g12190), S-locus glycoprotein (At5g39370), unknown protein (At1g53260)	AT1G61400: lectin protein kinase	EM
AT1G61390	2	2496	93.0	8.34	Pentatricopeptide (PPR) repeat-containing protein (At5g52850, At1g56690, At3g475390)	AT1G61360: Lectin protein kinase	EM
AT1G61400	0	2502	91.9	7.79	PPR repeat-containing protein (At5g27110, At3g47840), PSF2 (At3g12530), ATP-dependent DNA helicase (At4g25120)	AT1G61430, AT4G32300: Lectin protein kinase	EM
AT1G61380	0	2743	89.8	7.96	(EF-TU Receptor (At5g20480), Flagellin sensitive 2 gene (At5g46330), pectinesterase family protein (At2g26440), LRR kinase (At1g07650), ammonium transporter 2 (At2g38290)	AT1G61370: Lectin protein kinase; AT4G01570:Tetratricopeptide repeat (TPR)-like superfamily protein	EM
AT1G61370	0	2445	90.8	5.02	LecRLK (At1g61360), BRII - associated receptor kinase (At4g33430), B120 (At4g21390)	AT1G61380: Lectin kinase protein	EM
AT1G11280	4	2699	92.5	8.11	Ankyrin repeat family protein (At5g5860), RPL16A (At2g42740), mitochondrial transcription termination factor (At1g62150)	AT5G37500:K+ Channel; AT1G11350: Lectin kinase protein; AT2G33060: LRR-RLK; AT3G22150:TPR-like superfamily protein; AT5G55740:Chlororespiratory reduction	EM
AT4G11900	0	2579	95.8	7.21	Not Reported	AT1G28340: receptor like protein; AT2G17525:PPR superfamily protein	EM
AT1G61360	2	2647	90.9	6.39	Cysteine rich RLK (At4g23190), LecRLK (At1g61370), ATEP3 (At3g54420), Mildew resistance locus (At1g61560), Chitinase (At1g02360)	AT1G61390; AT4G32300: lectin kinase protein	EM
AT1G11330	2	2865	94.3	8.01	MAPKK (At4g29810), MEK1 (At4g26070), oxidoreductase (At3g28480)	AT1G11410: lectin kinase protein	EM
AT1G67520	0	2387	66.5	7.09	Not reported	AT1G64210: LRR RLK	EM, Cp ^a

Table 2 continued

Gene locus	SF	CDS	Mol. Mass (kDa)	pI	Co-expressed genes	Interacting partners	Localization
AT4G27300	0	2448	91.8	6.89	Receptor like protein (At2g15080), alcohol dehydrogenase (At1g64710), legume lectin family protein (At3g16530), oxidoreductase family protein (At1g34200)	Not reported	Pd
AT4G27290	0	2352	88.9	5.43	Phosphatase family protein (At2g37440), UV-B light-insensitive protein (At5g59930), unknown protein (At5g48175)	AT3G15950: TSK-associating protein 1	EM
AT1G61610	0	2529	95.3	6.61	Terpene synthase (At1g61120), CYP82G1 (At3g25180), Acyl activating enzyme 12 (At1g65890)	AT3G16030: LecRLK, AT3G23020:TPR-like superfamily protein	EM
AT3G16030	0	2553	96.2	7.52	Not Reported	AT1G61610, AT4G21380: lectin protein kinase; AT3G49170: Embryo-defective; AT5G16860:TPR-like superfamily protein	EM
AT1G61460	0	2585	67.6	8.31	LecRLK (At1g70130), protein kinase (At4g11480), DNA binding/transcription factor (At3g23220)	AT5G54710: Ankyrin repeat family protein; AT1G09900, AT4G18520, AT5G14820: PPR-like superfamily protein	NR
AT1G11350	0	2680	93.2	7.5	Resistance to Leptosphaeria maculans (At4g16990), TIR-NBS-LRR class protein (At5g46450), Garesponsive protein (At1g22690), GBF's Pro-Rich region interacting factor 1 (At2g20570)	Not reported	PM
AT1G11300	0	5104	184.	8.14	Methyltransferase (At3g21950), hydroxyproline-rich glycoprotein family protein (At4g28160), TCP family transcription factor 4 (At3g15030)	Not reported	EM
AT4G21380	0	2983	96.4	7.67	Pectinesterase family protein (At2g26440), Receptor like protein kinase (At5g60900), receptor like protein (At3g05650)	AT1G65790, AT3G16030: LecRLK; AT5G16860:TPR-like superfamily protein; AT3G49170: embryo defective	PM ^a , Pd, Vc ^a
AT4G03230	0	3258	113.	6.02	AME3 (At4g32660), SKP1 interacting partner 3 (At2g02350), unknown protein (At2g26470)	AT2G19130:lectin kinase protein: AT4G21370: S locus receptor kinase	EM
AT2G19130	0	2594	92.3	6.80	Protein kinase (At5g38210), DNA-3-methyladenine glycosylase (At3g12040), pEARL14 (At2g20960), ankyrin protein kinase (At2g43850), unknown protein (At5g02630)	AT3G02850:K + channel; AT4G03230:LecRLK; AT4G2137:S locus receptor kinase; AT1G05320: unknown function; AT1G12700: RNA processing factor	EM
AT1G65800	0	2781	95.9	8.36	NR	AT1G71400: CLAVATA2 related gene	Cy
AT1G65790	0	2532	95.9	8.41	NR	AT4G21380: LecRLK	PPI
AT4G21390	0	2762	95.1	6.85	Zinc finger family protein (At2g37430), Crinkly4 related protein (At3g55950), GNAT family protein (At2g32030), metacaspase 8 (At1g16420), LecRLK (At1g61370), unknown protein (At4g25030)	AT5G08490: TPR-like superfamily protein	PM ^a
AT4G32300	0	3121	89.7	6.45	NR	AT1G61360, AT1G61400: LecRLK	PM ^a
AT1G11410	0	2686	95.8	8.04	K + transporter (At4g22200), phosphoinositide binding (At5g43870), lipase (At1g75880), alcohol oxidase-related protein (At3g23410), glycosyl hydrolase protein (At3g19620), AOP1 (At4g0370), serine carboxypeptidase like protein (At2g27920)	AT1G11330: LecRLK	EM
AT1G11340	0	2706	102	7.48	HT1 (At1g62400), defectively organized tributaries DOT3 (At5g10250), unknown protein (At2g21080)		EM

Abbreviations: Cy cytosol, EM endomembrane system, PM plasma membrane, Pd plasmodesma, PPI pollen pistil interaction, Vc vacuole, SF spliced form, CDS coding sequence

^a Refers to experimentally observed localization of protein that was reported by SUBA database

Table 3 *Oryzae sativum* L type Lectin receptor-like kinase

Locus ID	SF	CDS	Mol. mass (kDa)	pI	Localization	Orthologous genes
LOC_Os07g15940	0	984	36.22	9.7	PM ^a , Cp ^b	Not reported
LOC_Os09g09510	0	2064	75.75	6.5	PM ^{a,b}	Not reported
LOC_Os05g13770	0	2349	86.74	8.4	PM ^{a,b}	Not reported
LOC_Os07g15930	0	1722	63.04	7	PM ^{a,b} , Ec ^a	Not reported
LOC_Os07g04040	0	2013	74.24	6.7	PM ^a , Cp ^b	Not reported
LOC_Os03g56180	0	1767	63.09	5.8	Cp ^a , PM ^a , Ec ^b	Not reported
LOC_Os07g03870	0	2283	84.11	7.2	PM ^a , Ly ^a , Cy ^b	Not reported
LOC_Os04g44910	0	2211	79.79	6.8	PM ^a , Mt ^b	GRMZM2G168953, Sb06g023520, AT2G37710
LOC_Os07g04130	0	2040	73.34	6.8	PM ^{a,b} , Cp ^a	Not reported
LOC_Os08g03100	0	1959	71.88	6.4	PM ^{a,b}	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os08g02996	2	2253	81.22	5.4	PM ^a	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os02g48210	0	2244	80.87	6.4	PM ^{a,b} , Vc ^a	GRMZM2G022514, GRMZM2G466298
LOC_Os06g22290	0	2295	80.98	6.7	Ec ^a , PM ^a , Cp ^b	GRMZM2G154766, Sb07g027400
LOC_Os08g03150	0	2109	76.77	6.8	PM ^a	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os04g01874	0	2196	78.87	5.9	PM ^a , Cp ^b	Not reported
LOC_Os06g17490	0	2097	73.63	6.8	Cp ^{a,b,c} , PM ^a	GRMZM2G159953, Sb10g010530, AT5G55830
LOC_Os09g16540	0	2172	79.87	6.4	PM ^a , Cp ^{a,b} , Vc ^b	GRMZM2G399087, Sb02g022340
LOC_Os07g38250	0	1767	63.24	6.3	Cp ^a , ER ^a	Not reported
LOC_Os01g57100	0	2091	74.56	5.5	Cp ^{a,c} , Ec ^b	GRMZM2G060271, GRMZM5G869971, Sb03g036370
LOC_Os04g22120	0	2241	80.14	5.9	PM ^a , Cp ^b	GRMZM2G474153
LOC_Os08g03240	0	2187	78.11	5.4	PM ^a , Vc ^b	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os10g30530	0	2076	74.21	6.8	Cp ^a , PM ^a	GRMZM5G820163, Sb01g020960
LOC_Os09g07730	0	1959	69.71	6.8	PM ^a , b, Cp ^{a,b,c}	Sb03g013430
LOC_Os04g01890	0	1998	73.87	8.5	PM ^a	Not reported
LOC_Os05g03450	0	2316	79.91	5.9	Cp ^a , PM ^a , Ec ^b	GRMZM2G351937, GRMZM2G381086, Sb09g002240, AT3G53380, AT5G03140
LOC_Os02g48200	0	2205	79.02	6.9	Cp ^a , PM ^b	GRMZM2G146866, Sb06g026570, AT5G06740
LOC_Os08g40270	0	2220	78.94	7.3	Cp ^{a,b,c}	GRMZM2G465165, Sb07g027410
LOC_Os08g40280	0	2202	77.97	5.9	Cp ^{a,b,c} , PM ^a	GRMZM2G154766, Sb07g027400
LOC_Os08g03150	0	2109	76.77	6.8	PM ^a	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os02g19530	0	2052	75.3	8.5	Cp ^a	Not reported
LOC_Os02g19550	0	1776	65.29	10	PM ^a	Not reported
LOC_Os02g26160	2	2088	74.13	9.7	PM ^a , Cp ^b	GRMZM5G816031, Sb04g009910, AT5G42120
LOC_Os02g42780	0	2025	73.24	7	PM ^{a,b}	Sb04g033790
LOC_Os03g15250	0	3936	143.8	6.8	PM ^{a,a} , Cp ^{a,c}	GRMZM2G400694, Sb10g011170
LOC_Os03g56160	0	1431	51.72	9.5	Cp ^a , PM ^a , Cy ^b	Not reported
LOC_Os03g60810	0	2106	75.11	5.6	PM ^{a,b} , Cp ^a	GRMZM2G093210, Sb01g003030
LOC_Os04g44900	0	2022	73.87	8	Cp ^a , PM ^{a,b}	GRMZM2G468756, Sb06g023510, AT2G37710, AT3G53810, AT4G02410, AT4G02420
LOC_Os06g10790	0	2133	75.45	6.3	PM ^{a,b} , Cp ^{a,b}	GRMZM2G349243, Sb03g031870
LOC_Os06g14260	0	2169	77.58	7.2	Cy ^a , Ec ^b	Not reported
LOC_Os07g03780	0	2139	78.65	6.3	PM ^{a,b}	Not reported
LOC_Os07g03790	0	1971	72.45	6.9	Cp ^a	Not reported

Table 3 continued

Locus ID	SF	CDS	Mol. mass (kDa)	pI	Localization	Orthologous genes
LOC_Os07g03810	0	2046	74.39	7.7	PM ^a , Cp ^b	Sb10g011010
LOC_Os07g03820	0	2097	76.02	6.8	PM ^{a,b}	Sb10g011000
LOC_Os07g03830	0	2016	74.13	6.7	PM ^{a,b}	Sb02g002240
LOC_Os07g03840	0	2097	76.35	6.8	PM ^a	Not reported
LOC_Os07g03860	0	1965	72.45	6.3	PM ^a , Cy ^b	GRMZM2G400720
LOC_Os07g03880	0	2079	77.58	7.7	PM ^a	Not reported
LOC_Os07g03890	0	3825	142.9	6.7	PM ^a	GRMZM2G397889, Sb10g011150
LOC_Os07g03900	0	2031	74.9	7.6	PM ^a , Vc ^b	GRMZM2G337768, Sb09g028880, Sb10g011160
LOC_Os07g03920	0	2829	103.9	6.9	PM ^{a,b}	Not reported
LOC_Os07g03930	0	1944	72.48	7.8	PM ^{a,b}	Not reported
LOC_Os07g03950	0	1911	70.71	8.2	PM ^{a,b}	GRMZM2G337768, Sb09g028880, Sb10g011160
LOC_Os07g03970	0	2022	74.79	7.2	PM ^{a,b}	Not reported
LOC_Os07g04000	0	2157	80.07	7.3	PM ^{a,b}	Not reported
LOC_Os07g04010	0	2010	74.54	8	PM ^a , ER ^b	Not reported
LOC_Os07g18230	0	2010	74.57	7.4	PM ^{a,b}	GRMZM2G469175, Sb02g009160
LOC_Os07g18240	0	2100	77.28	7.4	PM ^a , Cp ^{a,b}	GRMZM2G438871, Sb02g009170
LOC_Os07g38800	0	2094	75.71	7.9	PM ^{a,b} , Cp ^a	GRMZM2G443843, Sb02g037290
LOC_Os07g38810	0	2016	72.84	8	PM ^a	GRMZM2G443829, Sb02g037300
LOC_Os07g38820	0	2136	78.16	7.6	PM ^a , Cp ^b	Not reported
LOC_Os08g03002	0	1458	53.42	6.2	PM ^{a,b}	Not reported
LOC_Os08g03020	0	2031	73.9	7.6	PM ^{a,b}	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os08g03070	0	1539	55.85	6.3	PM ^{a,b} , Cp ^{a,c}	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os08g03090	0	1899	69.65	5.7	PM ^{a,b}	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os08g03160	0	2097	76.21	6.7	PM ^{a,b}	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os08g03170	0	1623	59.76	6.5	PM ^a	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os08g03240	0	2187	78.11	5.4	PM ^a , Vc ^b	GRMZM2G056109, Sb01g006670, Sb10g023375, AT5G10530, AT5G65600
LOC_Os09g09500	0	2007	74.91	8.3	Cp ^a , PM ^{a,b}	Not reported
LOC_Os10g30530	0	2076	74.21	6.8	PM ^a , Cp ^a	GRMZM5G820163, Sb01g020960
LOC_Os10g30540	0	2124	76.29	6.6	Cp ^a , PM ^b	GRMZM2G079376, GRMZM5G896540, Sb01g020950
LOC_Os10g38960	0	2025	71.99	6.8	Cp ^a , Cy ^a , PM ^b	Sb01g030300, AT3G08870, AT5G01540, AT5G01550, AT5G01560
LOC_Os12g38610	0	2061	72.69	7.4	NR ^a , PM ^b	GRMZM2G093736, Sb08g019050, AT3G55550

Abbreviations: Cp chloroplast, Cy cytosol, Ec extracellular, Ly Lysosome, PM plasma membrane, Vc vacuole, SF spliced form, CDS coding sequence

^a Localization prediction by CELLO v.2.5 (<http://cello.life.nctu.edu.tw/>)

^b Localization prediction by pSORT (<http://wolfsort.org/>)

^c Chloroplast localization signal confirmed by ChloroP (<http://www.cbs.dtu.dk/services/ChloroP/>)

arabidopsis L type (Table 1), 4 in arabidopsis G type (Table 2), 2 in rice L type (Table 3), 6 in rice G type (Table 4)}, had introns. We also analyzed the co-expression data as well as predicted interacting partners of the proteins of LecRLK family in arabidopsis and found many common

genes as interacting partners of LecRLK proteins. The main interacting partners found for LecRLK gene family were RLKs (LRR and LecRLK), K⁺ channel involved in stomatal movements and plant transpiration, proteins involved in embryogenesis and organelle biogenesis, brassinosteroid

Table 4 *Oryzae sativum* G type Lectin receptor-like kinase

Locus ID	SF	CDS	Mol. mass (kDa)	pI	Localization	Orthologous genes
LOC_Os01g66640	0	2373	87.8	7.0	PM ^{a,b}	Not reported
LOC_Os06g37690	0	2421	89.9	5.6	PM ^{a,b}	GRMZM2G418956, Sb10g022720, Sb10g022730
LOC_Os06g37670	0	2475	91.7	6.4	PM ^{a,b}	GRMZM2G418956, Sb10g022720, Sb10g022730
LOC_Os07g08860	0	2514	92.1	7.3	PM ^{a,b}	GRMZM2G420882, Sb09g001740
LOC_Os11g03880	0	2484	91.4	6.7	PM ^a , Cp ^b	Sb08g008420
LOC_Os12g03690	0	2487	91.6	6.7	PM ^a , Cp ^{a,b}	Sb08g008420
LOC_Os08g13870	0	2373	86.8	7.0	PM ^{a,b}	GRMZM2G309512, Sb03g046880
LOC_Os06g40030	0	2442	91.6	7.1	PM ^{a,b}	GRMZM2G418956, Sb10g022720, Sb10g022730
LOC_Os05g07470	0	1647	59.9	7.1	Ec ^a , Cp ^{b,c} , Mt ^b	Not reported
LOC_Os05g07080	0	2160	79.8	7.2	PM ^a , Cp ^b	Not reported
LOC_Os06g06920	0	2433	91.4	8.7	PM ^{a,b}	GRMZM2G099324, Sb10g004530
LOC_Os03g30890	0	2517	96.5	7.2	PM ^a , Vc ^b	Sb01g032150
LOC_Os08g13420	0	2514	91.9	7.1	PM ^{a,b}	GRMZM2G309512, Sb03g046880
LOC_Os01g45520	0	2232	82.3	8.1	PM ^{a,b}	Not reported
LOC_Os12g34320	2	2475	91.6	7.0	PM ^{a,b}	Not reported
LOC_Os06g37750	0	2409	90.9	6.0	PM ^{a,b}	GRMZM2G418956, Sb10g022720, Sb10g022730
LOC_Os01g47840	0	2346	87.7	6.5	PM ^{a,b}	GRMZM2G142207, Sb03g004860, Sb03g030710, Sb03g030720, Sb08g004830
LOC_Os02g13965	0	2511	93.7	6.5	PM ^a , Cp ^b	GRMZM2G383881, Sb04g008440
LOC_Os12g03670	0	2511	92.9	6.3	PM ^{a,b}	Not reported
LOC_Os12g03650	0	2508	92.7	6.1	PM ^{a,b}	Not reported
LOC_Os11g03840	0	2526	93.7	6.3	PM ^{a,b}	Not reported
LOC_Os11g03820	0	3804	146.	8.6	PM ^{a,b}	GRMZM2G328785, GRMZM2G366411, Sb09g008070, Sb09g017130
LOC_Os01g48020	0	2337	87.5	6.0	PM ^a	Not reported
LOC_Os06g06940	0	1974	73.1	7.5	PM ^{a,b}	Not reported
LOC_Os12g07800	0	2343	87.9	7.9	PM ^{a,b} , Ec ^a	GRMZM2G142207, Sb03g004860, Sb03g030710, Sb03g030720, Sb08g004830
LOC_Os01g47820	0	2394	90.2	6.5	PM ^{a,b}	GRMZM2G142207, Sb03g004860, Sb03g030710, Sb03g030720, Sb08g004830
LOC_Os01g48040	0	2319	86.9	6.6	PM ^{a,b}	GRMZM2G142207, Sb03g004860, Sb03g030710, Sb03g030720, Sb08g004830
LOC_Os06g06930	0	2457	92.9	6.9	PM ^{a,b}	GRMZM2G099324, Sb10g004530
LOC_Os12g03640	0	2190	83.8	7.2	PM ^{a,b}	GRMZM2G328785, GRMZM2G366411, Sb09g008070, Sb09g017130
LOC_Os01g48000	0	2445	89.4	6.6	PM ^{a,b} , Ec ^a	Sb03g004850, Sb03g006760, Sb03g030730
LOC_Os04g56080	0	2547	94.1	5.9	Ec ^{a,b}	Not reported
LOC_Os01g47810	0	2388	89.7	6.4	PM ^a , Ec ^a	GRMZM2G142207, Sb03g004860, Sb03g030710, Sb03g030720, Sb08g004830
LOC_Os06g29810	3	2538	92.1	6.5	Ec ^a , PM ^{a,b}	GRMZM2G123604, Sb10g014200, AT4G32300
LOC_Os04g34330	0	2397	87.8	6.3	PM ^{a,b}	GRMZM2G098037, GRMZM2G471596, Sb06g016350
LOC_Os04g34250	0	2424	89	6.9	PM ^{a,b}	Not reported
LOC_Os01g65010	0	2475	90.6	7.3	PM ^a , Cp ^{b,c}	Not reported
LOC_Os01g06240	0	2496	91.4	6.8	Ec ^a	GRMZM2G401606, Sb03g005540
LOC_Os01g12430	0	2418	87.8	5.1	PM ^{a,b}	GRMZM2G127650, Sb03g001300
LOC_Os05g07420	0	2412	88.0	6.9	PM ^a	Not reported
LOC_Os05g07450	0	2415	88.3	7.3	PM	Not reported
LOC_Os01g66610	0	2481	89.3	8.4	PM ^a , Cp ^{b, c}	GRMZM2G418689, Sb03g042320
LOC_Os03g62180	3	2577	91.7	7.6	Ec ^a , PM ^{a,b}	GRMZM2G113158, Sb01g001980, AT1G34300
LOC_Os05g34390	0	2466	88.8	10.	PM ^a , Mt ^a , Cp ^{a,b}	Not reported

Table 4 continued

Locus ID	SF	CDS	Mol. mass (kDa)	pI	Localization	Orthologous genes
LOC_Os04g23700	0	2340	87.5	5.9	Nu ^a , PM ^b	Sb07g026780, Sb07g026790
LOC_Os01g10710	0	2538	93.9	5.9	PM ^a , Cp ^{b,c}	Not reported
LOC_Os04g23720	0	2469	91.9	7.0	PM ^{a,b}	Sb07g026780, Sb07g026790
LOC_Os04g23760	0	2592	95.8	6.7	PM ^{a,b}	Sb07g026780, Sb07g026790
LOC_Os04g12580	0	2346	87.2	4.9	Ec ^a	GRMZM2G017164, GRMZM2G303909, Sb06g002210, Sb06g019890, AT5G60900
LOC_Os04g23600	0	1980	74	7.7	PM ^a , Cp ^b	Not reported
LOC_Os04g34390	0	2409	88.5	6.5	PM ^{a,b}	GRMZM2G098037, GRMZM2G471596, Sb06g016350
LOC_Os04g34290	0	2451	91.1	5.2	PM ^{a,b}	Sb08g006890
LOC_Os01g12410	0	2289	85.0	6.6	PM ^{a,b}	Not reported
LOC_Os04g23620	0	2130	79.8	7.6	PM ^a , Cy ^b	Not reported
LOC_Os04g42740	0	2382	86.4	6.7	PM ^{a,b}	Not reported
LOC_Os04g28780	0	2454	90.3	6.5	PM ^{a,b}	Sb09g007250
LOC_Os04g34420	0	2427	89.1	6.3	PM ^{a,b}	GRMZM2G301124, GRMZM2G413030, Sb02g033290
LOC_Os04g34270	0	2055	75.0	6.9	PM ^a	Not reported
LOC_Os04g34300	0	2034	75.2	6.4	PM ^{a,b}	Not reported
LOC_Os04g34410	0	2484	91.7	7.5	PM ^a	Sb02g033290
LOC_Os10g20160	0	2424	89.2	5.7	PM ^a	Sb02g033290
LOC_Os11g10280	0	2373	88.1	6.2	PM ^a	Not reported
LOC_Os04g01320	0	2442	90.7	6.5	PM ^a	Sb09g007250
LOC_Os01g66680	0	2247	82.4	7.5	PM ^a , Ec ^a	Not reported
LOC_Os01g12290	0	2328	86.7	6.6	PM ^{a,b}	Sb03g001330, Bradi2g07310, GRMZM2G318864
LOC_Os01g10700	0	2253	84.5	5.4	Ec ^a , Vc ^b	Not reported
LOC_Os07g36780	0	2475	90.4	7.0	Ec ^a , PM ^{a,b}	GRMZM2G309025, Sb02g036040, AT2G19130
LOC_Os04g07200	0	1827	68.2	4.9	PM ^a , Nu ^{a,b}	Not reported
LOC_Os11g10260	0	3282	121	8.6	PM ^a , Mt ^a , Nu ^b	Not reported
LOC_Os01g12420	0	2442	88.5	6.2	PM ^{a,b}	GRMZM2G127650, Sb03g001300
LOC_Os09g28180	0	2697	94.7	7.6	PM ^a , Cp ^b	GRMZM2G009166, Sb02g026520
LOC_Os06g06960	0	2418	90.4	7.1	PM ^a , Vc ^b	GRMZM2G099324, Sb10g004530
LOC_Os05g42210	0	2514	90.8	6.2	PM ^a , Ec ^a	GRMZM2G075247, GRMZM2G106702, GRMZM2G322723, GRMZM2G465999, Sb01g016720, Sb02g035920, Sb03g036650, AT1G65790, AT1G65800, AT4G21380
LOC_Os03g35600	0	2607	95.8	8.0	PM ^a	GRMZM2G075247, GRMZM2G106702, GRMZM2G322723, GRMZM2G465999, Sb01g016720, Sb02g035920, Sb03g036650, AT1G65790, AT1G65800, AT4G21380
LOC_Os01g57560	0	2508	90.6	6.6	PM ^{a,b} , Ec ^a	GRMZM2G075247, GRMZM2G106702, GRMZM2G322723, GRMZM2G465999, Sb01g016720, Sb02g035920, Sb03g036650, AT1G65790, AT1G65800, AT4G21380
LOC_Os06g41510	0	2496	91.7	7.0	PM ^{a,b}	GRMZM2G334592, Sb03g006110, Sb03g040620
LOC_Os12g44320	0	2697	95.6	6.3	PM ^{a,b}	Sb08g023250
LOC_Os06g41560	0	2574	93.6	7.0	PM ^{a,b}	GRMZM2G334592, Sb03g006110, Sb03g040620
LOC_Os06g41500	0	1863	67	6.7	PM ^a , Vc ^b , Cp ^b	GRMZM2G334592, Sb03g006110, Sb03g040620
LOC_Os09g37800	0	2568	93.2	7	PM ^{a,b}	GRMZM2G353726, Sb02g032060, Sb02g032110
LOC_Os07g36544.2	2	2580	92.6	5.9	PM ^{a,b} , Ec ^a , ER ^b	GRMZM2G465999, Sb02g035920
LOC_Os01g73390	0	1680	58.6	7.4	PM ^a	GRMZM2G309512, Sb03g046880
LOC_Os04g39910	0	2385	88.6	5.1	PM ^{a,b} , G ^b	GRMZM2G017164, GRMZM2G303909, Sb06g002210, Sb06g019890, AT5G60900

Table 4 continued

Locus ID	SF	CDS	Mol. mass (kDa)	pI	Localization	Orthologous genes
LOC_Os04g01310	0	2487	91.6	6.4	PM ^a , Vc ^b	Sb09g007250
LOC_Os03g12150	0	2532	91.4	7.1	Cp ^a , PM ^{a,b}	GRMZM2G411668, Sb01g042550
LOC_Os05g07080	0	2160	79.8	7.2	PM ^a , Cp ^b	Sb09g023490
LOC_Os01g40499.2	2	2571	92	7.6	PM ^{a,b}	Not reported
LOC_Os04g12540	0	2814	104	6.5	Ec ^a , PM ^{a,b}	Not reported
LOC_Os02g52850	0	2718	95.6	7.4	PM ^{a,b}	GRMZM2G058220, GRMZM2G152793, Sb04g034250, AT5G35370

Abbreviations: Cp chloroplast, Ec extracellular, Mt mitochondria, Nu nucleus, PM plasma membrane, G golgi apparatus, SF spliced form, CDS coding sequence

^a Localization prediction by CELLO v.2.5 (<http://cello.life.nctu.edu.tw/>)

^b Localization prediction by pSORT (<http://wolfpsort.org/>)

^c Chloroplast localization signal confirmed by ChloroP (<http://www.cbs.dtu.dk/services/ChloroP/>)

mediated signalling pathway genes and Ankyrin repeat family protein which are involved in disease resistance and antioxidation metabolism (Tables 1, 2). We also found RLKs (LRR and CRLKs), brassinosteroid receptor kinases, K⁺ transporters, ankyrin repeat family proteins along with various proteins involved in disease resistance, defence response and MAP kinase signalling pathway proteins etc. as co-expressed genes (Tables 1, 2).

Chromosomal localization of LecRLK gene family

To visualize the genome organization of LecRLK gene family members, chromosomal -localization maps were constructed (<http://arabidopsis.org/jsp/ChromosomeMap/tool.jsp>) for Arabidopsis and (<http://viewer.shigen.info/oryzavw/maptool/MapTool.do>) for rice. Twenty-three out of 32 Arabidopsis G type LecRLKs were found to be localized on chromosome 1 while Arabidopsis L type genes were scattered mainly over chromosome 3 (14 genes) and chromosome 5 (16 genes) (Fig. 1). In the case of rice G type LecRLKs, the genes were mainly found on chromosome 1 and 4 while L type rice LecRLK was found mainly on chromosome number 7 (Fig. 2). In all the above cases, the creation of regions with high density of genes was probably a result of tandem duplication repeat of genes.

Phylogenetic analysis

We have constructed rooted phylogenetic trees which were bootstrapped with 1000 replicates using MEGA version 5 (Tamura et al. 2011). The phylogram of L type LecRLKs and G type LecRLKs was constructed separately and clustered into subgroups. As shown in Fig. 3a, the phylogram of L type LecRLK family of rice and arabidopsis subdivided into four sub-groups, with group IV being largest of them. Similarly, for G type LecRLKs (Fig. 3b),

the phylogram was sub-divided into four subgroups. The division of individual clades was supported by high bootstrap values. The genes with high sequence similarity and 100 bootstrap value were regarded as homologues of each other. Such gene pairs have been shown connected to each other in the cladogram (Fig. 3b). Also, the genes which were found to be highly similar to each other (<95 bootstrap value) and also were next/near neighbors of each other on chromosomal maps have been predicted as tandem duplication events. Such gene groups have been marked in shaded boxes in cladogram (Fig. 3a, b).

Analysis of conserved motifs

To analyze conserved motifs in LecRLK protein, MEME Suite-GLAM2 version 4.8.0 was used. (<http://meme.nbcr.net/meme/intro.html>). A number of conserved motifs were found in kinase domain (data not shown). This confirmed that the kinase domain has maintained its conservation during the course of evolution in both the species. Lectin domain, however, exhibited huge diversity in sequence and very few conserved motifs were found in this region of great diversity. Most conserved region in lectin domain of G type rice Lectin kinase was found to be in 96 out of 100 sequences. The motif structure was S[GS]xVLWQSFD[HY]PTDTLLPG[QA]K[LI][GT]W[NGD]Kx[LT][GV]. Similar motif (LW[QE]SF[DE]H[PL][GT][DN]T[LM]LPG[MS][KS]L[GM]Y[ND]) was found in 31 out of 32 arabidopsis G type LecRLK (Fig. 4a). Another conserved motif was found in rice as well as Arabidopsis G type lectin domain which was present in 76 out of 100 sequences in rice and 27 out of 32 sequences of Arabidopsis (Fig. 4b). A cysteine rich conserved motif within PAN domain, which is an additional domain present in most of G type LecRLKs, was also found (Fig. 4c). Similarly, motifs were found in L type LecRLK. These motifs were different from the ones

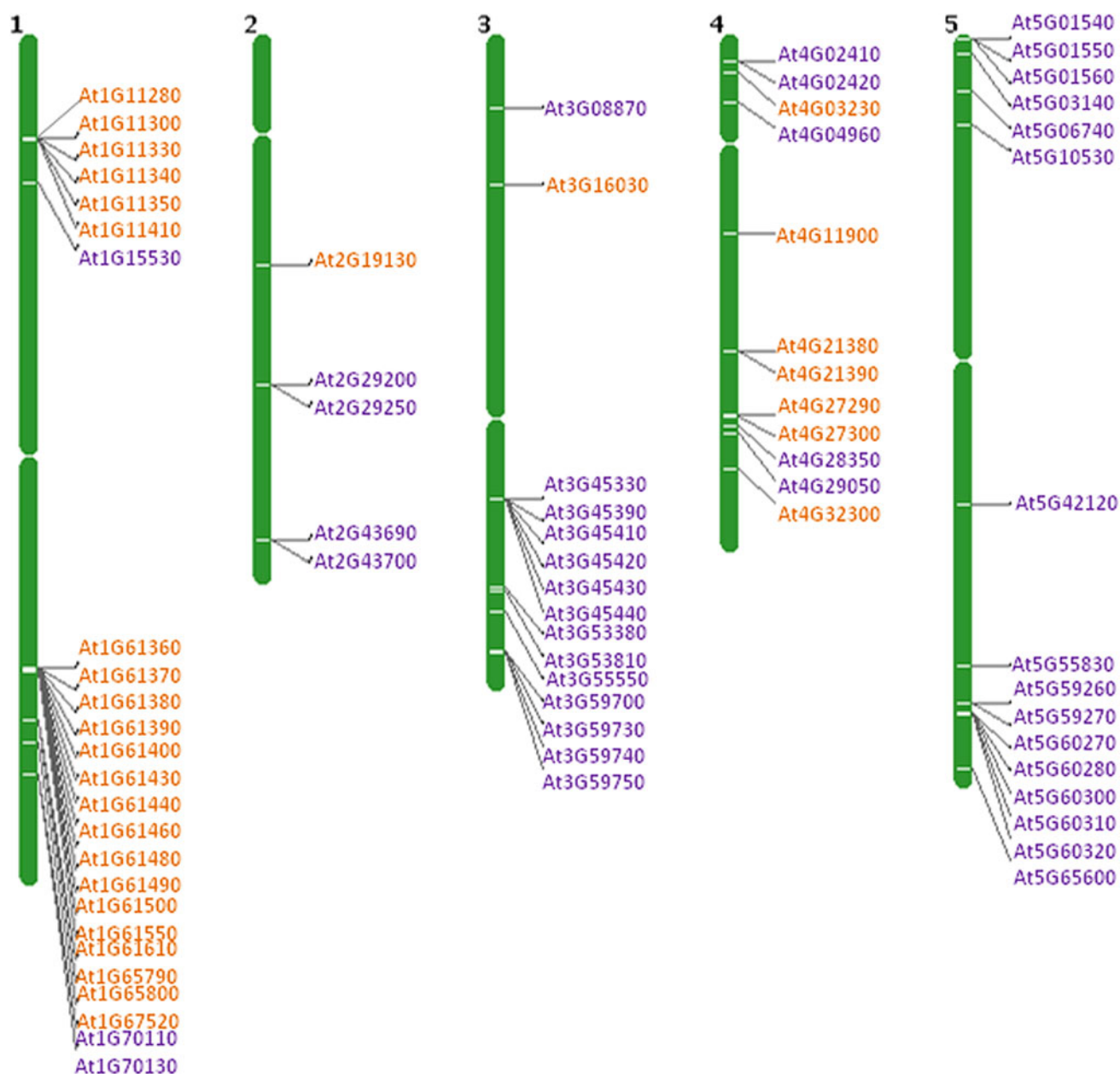


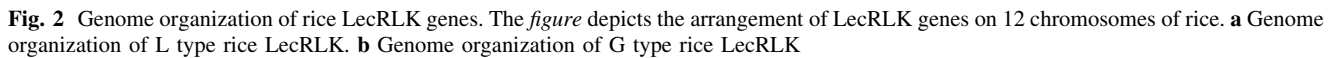
Fig. 1 Genome organization of Arabidopsis LecRLK genes. The figure depicts the arrangement of LecRLK genes on five chromosomes of Arabidopsis. L type lectin receptor-like kinase family genes

are depicted by purple colour while G type LecRLK genes are depicted by orange colour

found in G type LecRLKs but were conserved between L type LecRLKs in rice and Arabidopsis (Fig. 4d, e). To visualize the conserved motifs on LecRLK protein, structure models were constructed for lectin domain of L and G type LecRLKs. ESyPred3D Web Server 1.0 (<http://www.fundp.ac.be/sciences/biologie/urbm/bioinfo/esypred/>) predicted 2BQP chain 'A' as template for lectin domain of L type of LecRLK but it was unable to predict any homologue for lectin domain of G type LecRLK. Therefore, ab initio

method was used for the construction of the model for lectin domain of G type LecRLK.

The software used was i-Tasser (<http://zhanglab.ccmb.med.umich.edu/I-TASSER/>). PyMol was chosen as software for visualization and manipulation of structural models. Figure 5 shows homology modelling for L and G type lectin domains. Motifs derived in Fig. 4 are represented in different colour format on the structures. Figure 5a shows lectin domain of L type LecRLK. The motifs predicted in Fig. 4d



Gene expression analysis

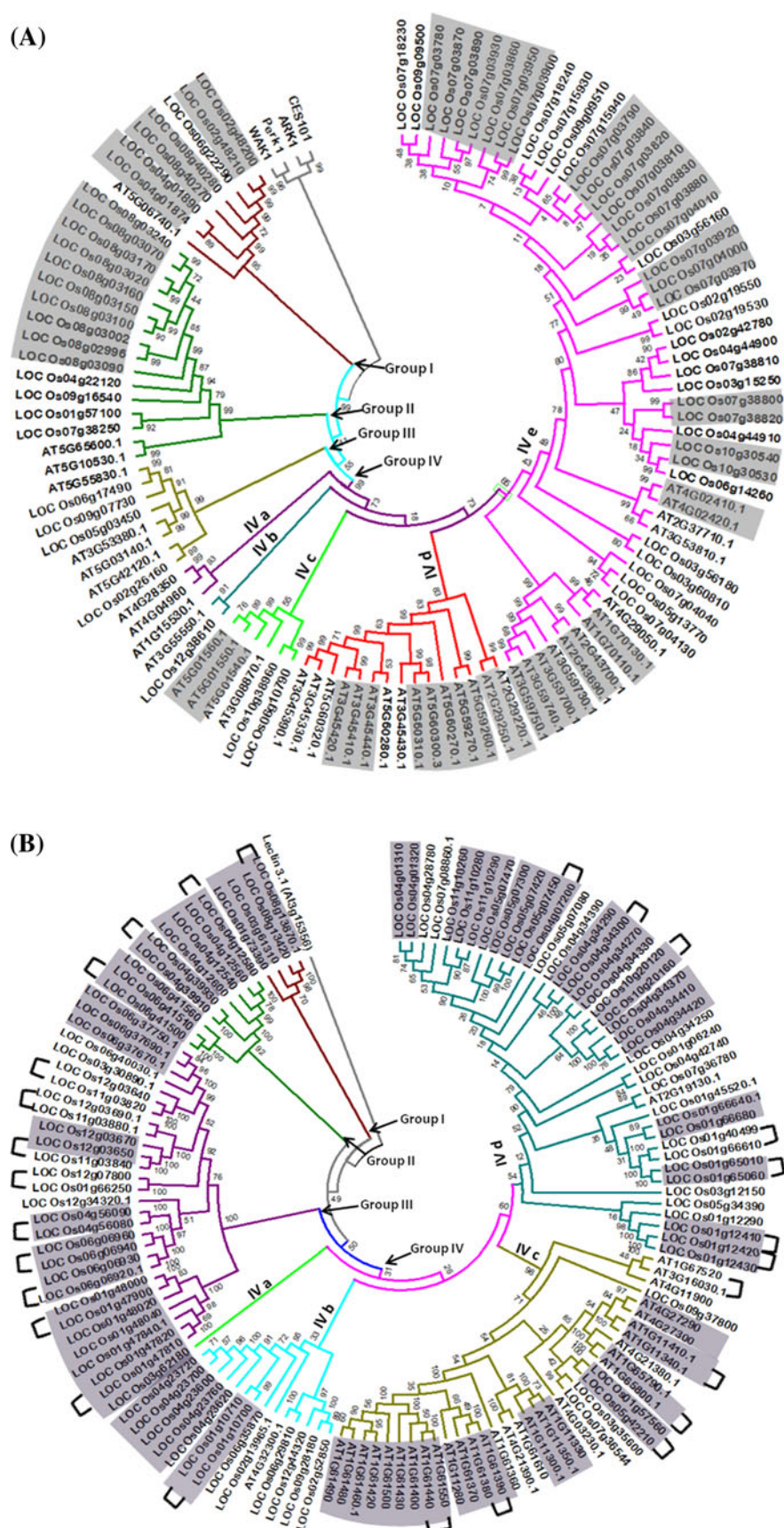
The rice G type LecRLK family is highly expressed in stress conditions (Fig. 8c). The effect is more pronounced in drought, salt stress, heat, cold and biotic stress and relatively less visible in hormonal treatment and anoxia. 43.75 % of rice L type LecRLKs showed visible differences in their transcript levels in the case of heat stress and are less affected by all the other stresses. Just like its rice

Discussion

LecRLKs are one of the members of plant receptor-like kinase family which are found in all higher plants. It has a characteristic extracellular domain which resembles lectin proteins. The N terminal region (i.e. the lectin domain) of LecRLK family members differentiates this group of proteins into three classes: L type, G type and C type family. The L type LecRLK family has resemblance with legume lectin family of proteins. They have a concanavalin-A like domain which comprises of a sandwich of 12–14 β strands in 2 sheets with complex topology. The legume lectins are largest family of lectin proteins and are differentiated amongst members on the basis of carbohydrate specificities. G type LecRLK, on the other hand, possess bulb type lectin domain (IPR001480), which has a β -barrel constituting 12 β strands. These types of lectin proteins have

Fig. 3 Phylogenetic analysis of LecRLK gene constructed using neighbour-joining method using 1000 bootstrap replicates.

Homologous gene pairs with 100 bootstrap value have been joined with connectors. Tandem duplication events have been marked in shaded boxes in cladogram **a** Phylogenetic tree of L type LecRLK gene family. **b** Phylogenetic tree of G type LecRLK gene family



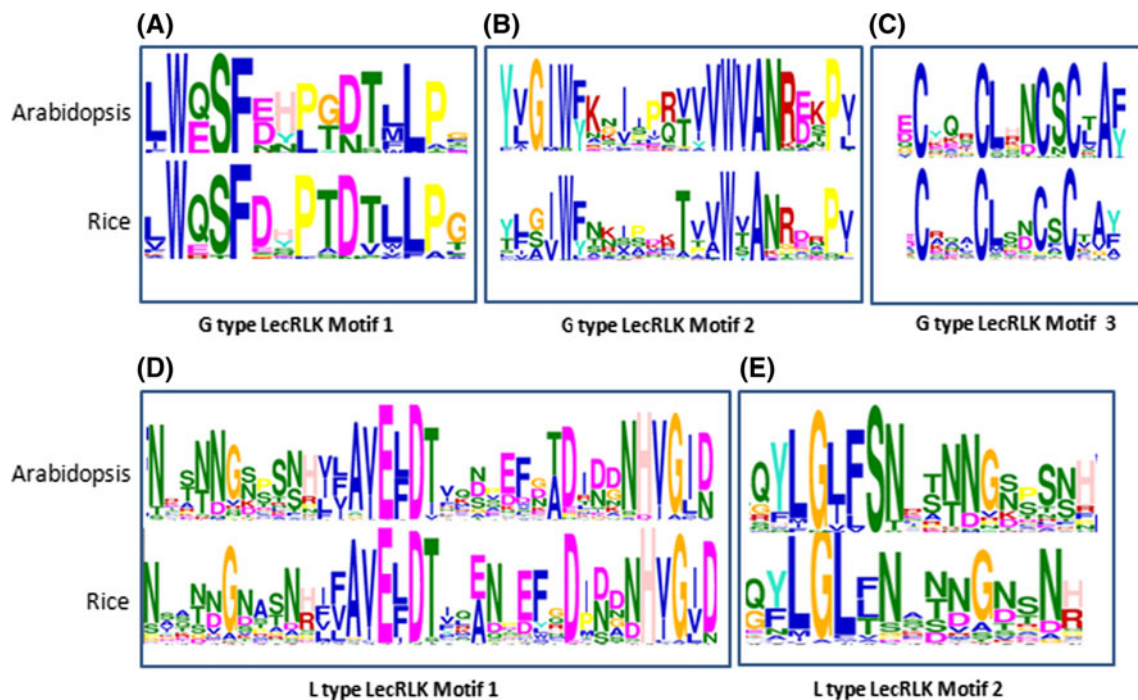
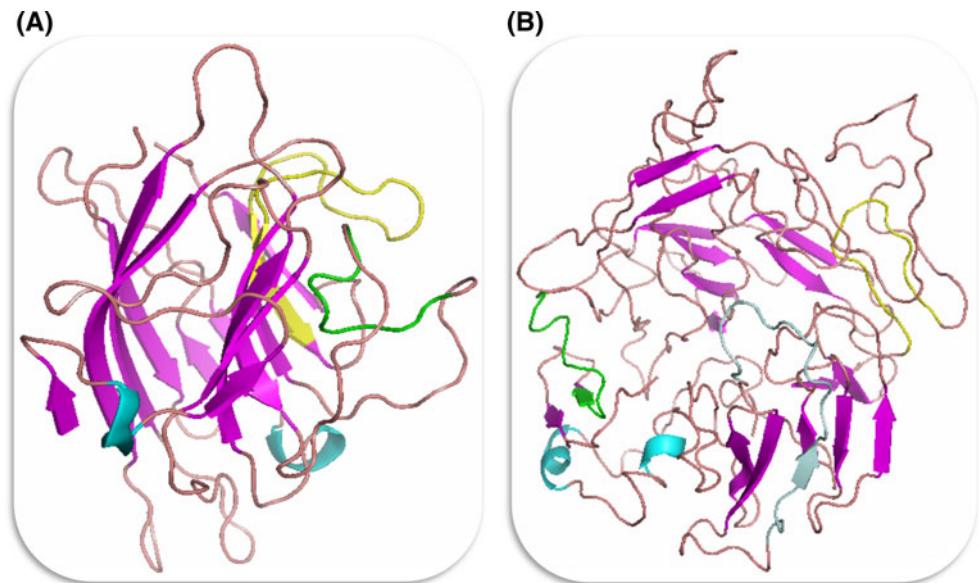


Fig. 4 Conserved motifs in lectin domain of G type and L type LecRLK proteins. *Uppers panel* of each motif is arabidopsis sequence while the *lower panel* is from rice sequence. Only the motifs that were found to be conserved both in rice and arabidopsis lectin domain were

chosen. **a–c** conserved motifs in lectin domain of G type LecRLK protein. **d–e** conserved motifs in lectin domain of L type LecRLK protein

Fig. 5 Structure models for L type (a) and G type (b) LecRLK. The sheets are colored dark pink; loops are colored pale pink and helix as bright blue. **a** homology model of L type LecRLK. The conserved motifs defined in Fig. 4d and e are defined here by yellow and green color respectively. **b** homology model of G type LecRLK. The motif defined in Fig. 4a, b and c are defined here in yellow, dull blue and green respectively



affinity towards mannose. Apart from lectin domain, G type LecRLKs have S-locus glycoprotein domain which was initially identified in brassicaceae and was found to be involved in recognition of self pollen to prevent inbreeding. The G type LecRLKs also possess a PAN2 domain, which is known to be involved in protein–protein or protein–carbohydrate interaction (Tordai et al. 1999; Loris 2002; Naithani et al. 2007).

We have systematically studied different groups of LecRLKs from rice and arabidopsis and compared the expression profile of LecRLKs from the two plants. To the best of our knowledge, complete analysis and a comparative expression profile of all classes of LecRLK family, described in this work, has not been previously reported. We have found 74 arabidopsis LecRLKs (42 L type and 32 G type) and 172 rice LecRLKs (72 L type and 100 G type).

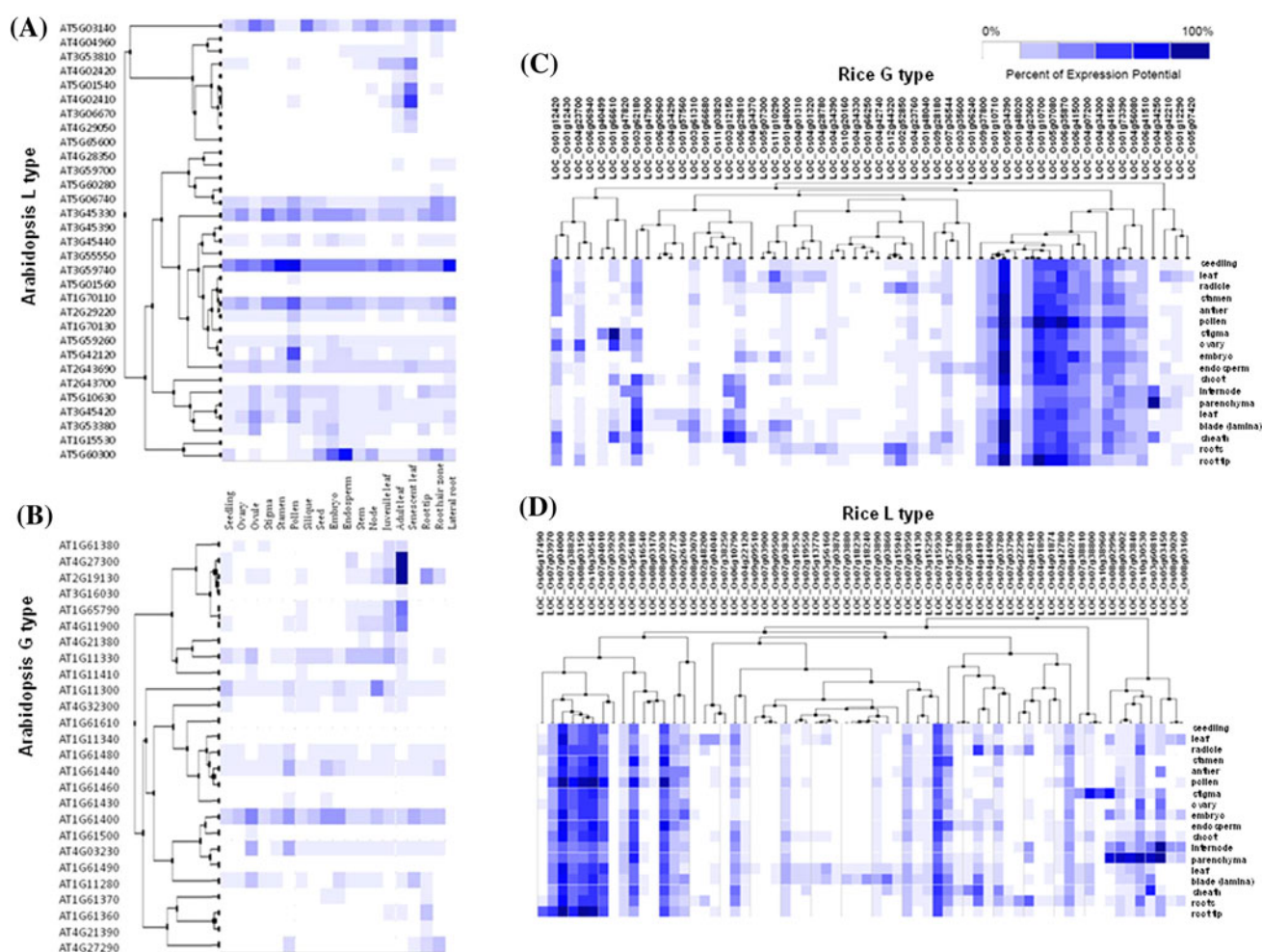


Fig. 6 Expression profile of arabidopsis and rice LecRLK gene family in different anatomical features depicted as heat map. **a** Expression profile of arabidopsis L type family in anatomy.

b Expression profile of arabidopsis G type family in anatomy. **c** Expression profile of rice G type family in anatomy. **d** Expression profile of rice L type family in anatomy

The initial analysis of physiological parameters suggests that the LecRLK family of proteins is very diverse, both in the size as well as charge. The gene family had a few similarities too. The whole family is rather intron-less, with only 13 genes out of 246 having intronic region. Introns are general characteristic of genes of higher eukaryotes. Alternative splicing is a mechanism which greatly enhances the biodiversity of proteins that genome of a particular organism can code for. In humans, ~95 % genes are alternatively spliced (Pan et al. 2008) whereas in Arabidopsis, about 5,846 genes, covering 21.7 % of Arabidopsis genome are intronless (Jain et al. 2008). According to a study carried out by Zou et al. (2011), while metazoans and other eukaryotes in a combination have about 50 % of eukaryote specific intronless genes, plants have none. Also, Yang et al. (2009), have stated that the genomes of rice, Arabidopsis and poplar are rich in species specific genes that are intronless. In humans, duplication events have been reported to be more common in intronless genes (Zou et al.

2011), although such study has not been carried out in plants.

The basic structure of LecRLK protein comprises of N terminal lectin domain, C terminal kinase domain and an intermediate transmembrane region. Due to the presence of this transmembrane region, the protein family is expected to be localized in plasma membrane or in general endo-membrane system. As predicted by the localization softwares used, LecRLK protein was majorly found to be localized in membrane, with a few exceptions, where protein was found to be localized in mitochondria or chloroplast. Arabidopsis G type LecRLK protein AT1G67520 has been experimentally proven to be plastid localized by GFP tagging as reported by SUBA (<http://suba.plantenergy.uwa.edu.au/>). Another receptor like kinase, PRO25, which is an EGF like domain containing receptor like kinase has been shown to interact with the amino terminus of a light-harvesting chlorophyll a/b-binding protein (LHCP), a chloroplast thylakoid membrane

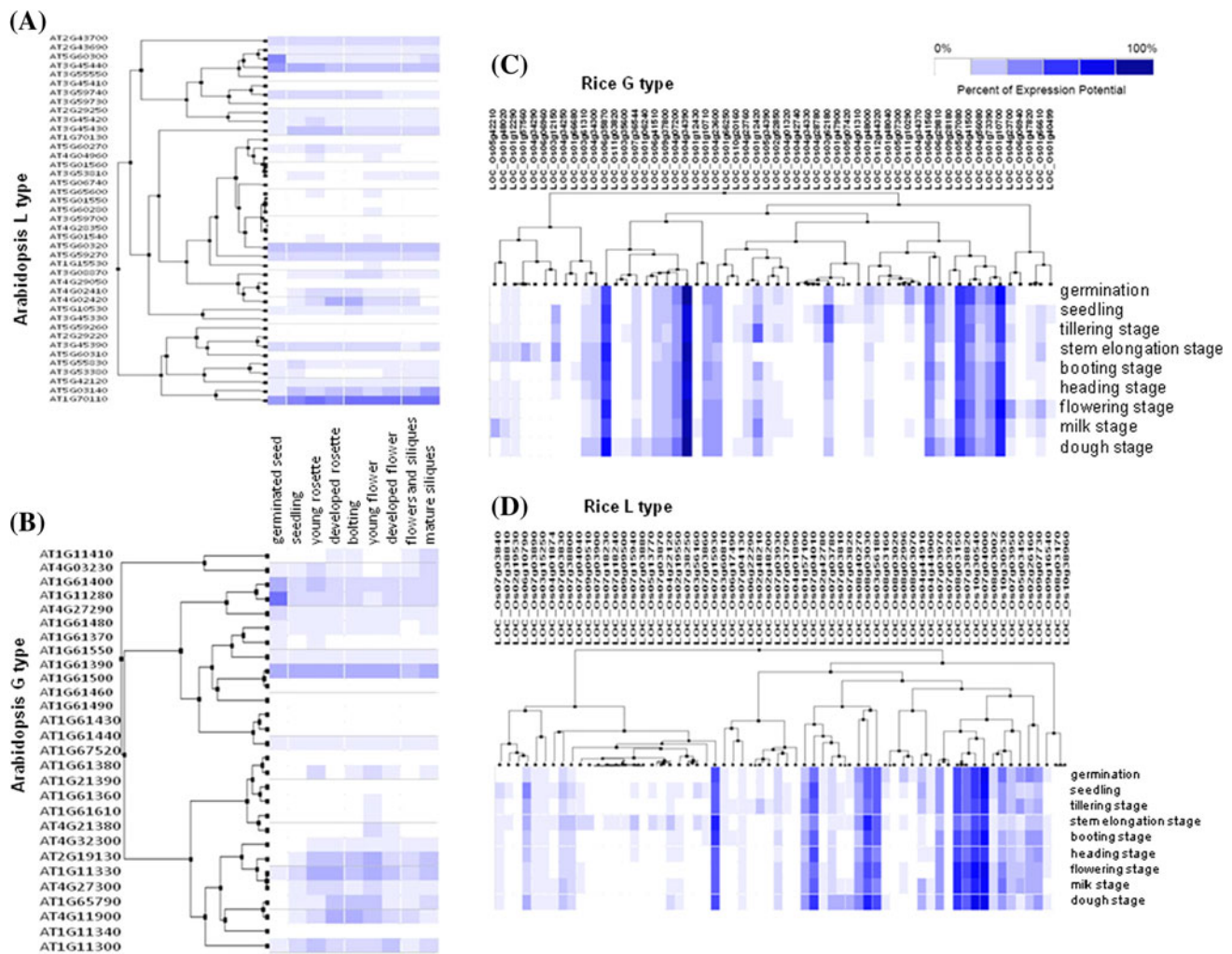


Fig. 7 Expression profile of arabidopsis and rice LecRLK gene family in different developmental conditions depicted as heat map. **a** Expression profile of arabidopsis L type family in different developmental stages. **b** Expression profile of arabidopsis G type

family in different developmental stages. **c** Expression profile of rice G type family in different developmental stages. **d** Expression profile of rice L type family in different developmental stages

protein (Kohorn et al. 1992). Therefore, it could be said that there is a high chance of few LecRLKs to be plastid localized though the function of these proteins in chloroplast is yet to be ascertained. It was observed that there were variations in prediction of localization of protein by different softwares, therefore, we emphasize the need of experimental verification of these predictions. The interacting partners data and co-expression data of the arabidopsis LecRLK protein also indicated that the gene family is involved in various functions ranging from signalling (brassinosteroid mediated signalling pathway genes, MAP kinase signalling pathway genes and interaction and coexpression with other RLKs) to stress response (ankyrin repeat family proteins, EF-TU Receptor, Chitinase, Mildew resistance locus) to plant physiology (Pentatricopeptide repeat superfamily protein associated with organelle biogenesis, gated outwardly-rectifying K^+ channel required

for stomatal movements and plant transpiration, proteins required for embryogenesis). The potential role that this gene family might be playing in physiology of plant is further strengthened by its putative interacting partners and co-expressed genes. The K^+ channel is required for regulation of osmotic balance and is essential for functioning of many cytosolic enzymes (Mahajan et al. 2008), while MAP kinase signalling pathway is an evolutionary conserved pathway that has many varied and well documented responses from cell division, differentiation and response to hormones and stress (Sinha et al. 2011; Bhardwaj et al. 2011).

Phylogenetic analysis

To understand evolutionary relationship between different subgroups of LecRLK, a rooted phylogram was constructed using *MEGA* version 5 with 1000 bootstrap replicates

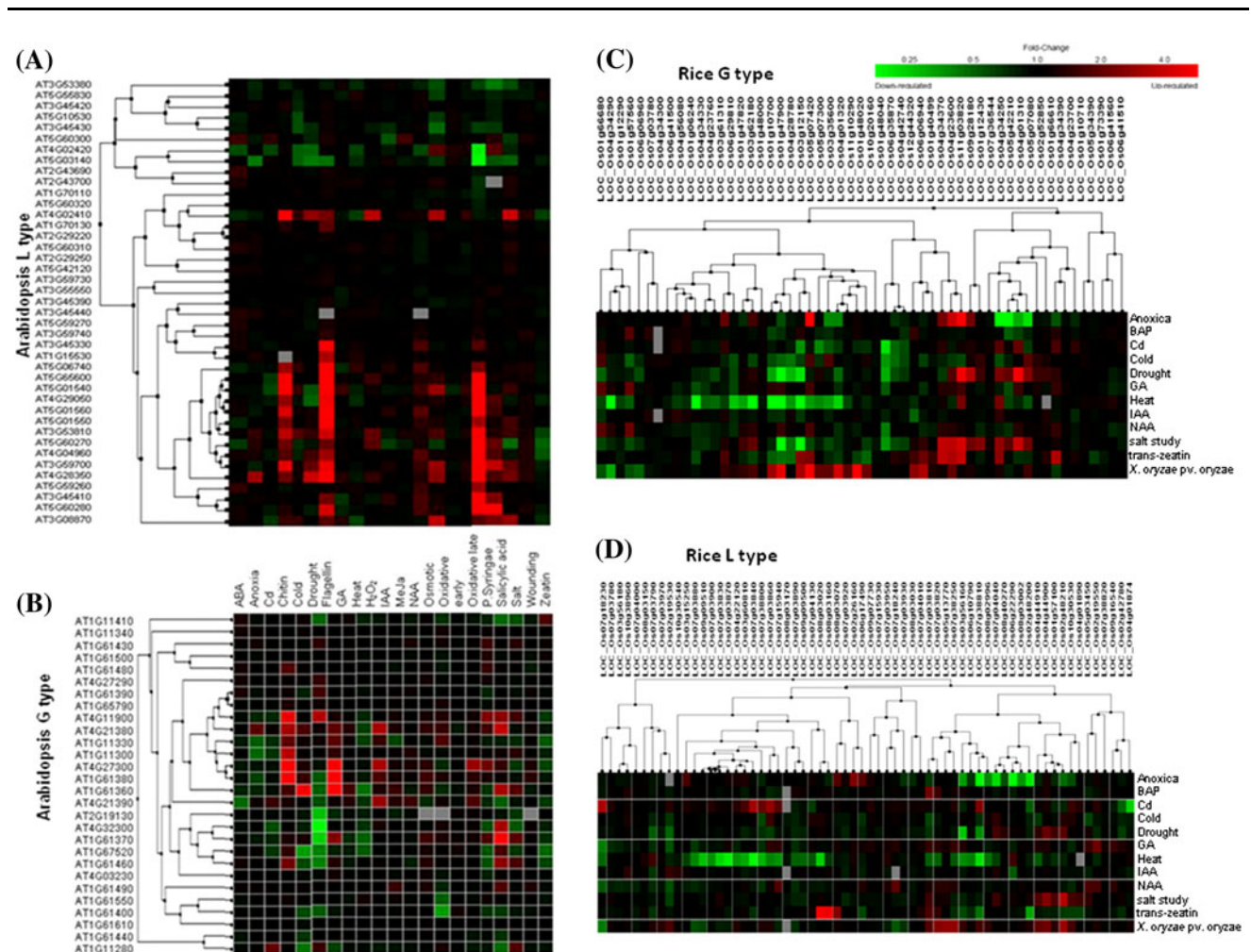


Fig. 8 Expression profile of arabidopsis and rice LecRLK gene family in different stress conditions depicted as heat map. **a** Expression profile of arabidopsis L type family in different stress conditions. **b** Expression profile of arabidopsis G type family in different stress

(Tamura et al. 2011). We found that L, G and C type LecRLKs of Arabidopsis and rice (data not shown) have evolved separately from each other. This was an expected outcome since the N terminal lectin domains of L type and G type LecRLKs have different origins. L type LecRLKs are similar to legume LecRLKs while G type LecRLKs have bulb lectins in lectin domain. Both types of lectins have different structure and carbohydrate affinities. Bootstrap values of the nodes was observed to be low for L type LecRLKs while the cladogram of G type LecRLKs had high bootstrap value of nodes. The lower bootstrap value of the nodes could be due to high sequence variation amongst individual clades (Shiu and Bleecker 2003). A stronger bootstrap support for individual clades added confidence to the cladogram construction. Also, using the bootstrap cladogram, we were able to derive homologous gene pairs for L and G type LecRLKs (Fig. 3a, b).

conditions. **c** Expression profile of rice G type family in different stress conditions. **d** Expression profile of rice L type family in different stress conditions

Upon close examination, we found that many of the subgroups contained *LecRLK* genes that lie very close to each other on chromosomal arrangement. To name a few, 12 genes mapped together on chromosome 1 of Arabidopsis (At1G61360 to At1G61550); five genes from Arabidopsis from At5g59260 to At5g60310; nine genes of rice from LOC_Os04g34250 to LOC_Os04g34410; ten genes of rice from LOC_Os08g02996-LOC_Os08g03240; 17 genes of rice from LOC_Os07g03780 to LOC_Os07g04010, are very closely related in sequence as depicted by the respective cladograms. Most of the genes that were found to be next/ near neighbours of each other on chromosomal organization were also found to be closely related to each other in cladogram with bootstrap value of >90. Therefore, keeping in consideration that the tandem duplicated genes are generally located within 30 Kb of each other or lie within 10 predicted open reading frames of a gene (Shiu and Bleecker 2003) and the support lend by the sequence similarity in cladogram,

Table 5 Percentage expression of LecRLK gene

Stress treatment	Rice		Arabidopsis	
	G type	L type	G type	L type
Anoxia	21.9	21.8	10.7	7.3
Cadmium	27.4	21.8	10.7	12.2
Cold	37.0	14.0	25	21.9
Drought	45.2	23.4	42.8	19.5
GA	27.4	25	3.6	4.8
Heat	41.1	43.7	17.8	12.1
Biotic stress	34.2	6.2	32.1	46.3
IAA	30.1	15.6	10.7	7.3
NAA	30.1	14.0	17.8	26.8
Salt stress	41.1	21.8	32.1	26.8
t-Zeatin	31.5	28.1	21.4	17.0

Numbers of genes that are expressed in a particular stress condition are expressed as percentage of total number of genes belonging to that particular gene group

we could strongly predict these genes to be tandem duplication events (Yu et al. 2005; Clauss and Mitchell-Olds 2003). The theory upon which the original models of gene duplication were based stated that upon duplication of a particular gene, one copy would resume its original function while the other would be able to perform a newer function (Ohno 1970; Babbitt et al. 2007). Therefore, these tandem duplications are not random and have arisen so that the genes with multiple functions could improve on one function without compromising on the other function of the gene (Babbitt et al. 2007). This theory has been proven in case of *Saccharomyces cerevisiae* galactose metabolism pathway (Hittinger and Carroll 2007).

Following this, we checked that expression pattern of the putatively duplicated genes was not similar. Similar results have been reported by Clauss and Mitchell-Olds 2003. Also, Bouwmeester and Govers 2009 have concluded that there was no correlation between the expression pattern of a particular gene and its position in the phylogenetic tree in case of L type LecRLK of Arabidopsis.

Conserved motif analysis

The search for conserved motif revealed signature motifs in G type as well as L type LecRLKs. The conserved motifs found in L type LecRLKs were highly conserved in both rice and Arabidopsis. Similar pattern was observed in conserved motifs of G type LecRLKs of Arabidopsis and rice as well. This shows that although the lectin domain has not maintained conservation in its sequence during evolution, a phenomenon that is responsible for the diversity of the gene family in recognition of signals and functions, certain signature motifs within lectin domain in the gene family have retained conservation. These motifs were visualized on

structural models of L and G type LecRLKs. In case of L type lectin domain, one of the conserved motifs was found near the carbohydrate binding site as predicted by Barre et al. 2002. The other conserved motif was not found to be part of any predicted domain. For lectin domain of G type LecRLKs too, the conserved motif was not found to be part of any predicted domain. Though, motif shown in Fig. 4a and coloured as yellow in Fig. 5b was found at the end of BULB type lectin domain profile. The second domain was not part of any domain found by the prosite scan.

Gene-expression analysis

We analyzed the expression of 74 arabidopsis and 172 rice genes from LecRLK family for various environmental factors (like biotic stress, abiotic stress and hormonal signals), different developmental stages and different anatomical features of plant. There have been very few functional studies carried out on the LecRLK family. We tried to correlate the results of available experimental evidences with the bioinformatic analysis. L type Arabidopsis LecRLK genes namely At3g59700 and At5g01540 have been reported to impart biotic stress tolerance towards *Pseudomonas syringae* (Desclos-Theveniau et al. 2012; Singh et al. 2012 respectively). These results can be correlated to our analysis as both At3g59700 and At5g01540 were highly upregulated in response to *P. syringae*, chitin and flagellin. Also, G type rice LecRLK Os6g29810 has been reported to confer resistance against fungal pathogen *Magnaporthe grisea* (Chen et al. 2006). Although our study on gene expression in rice did not include response to fungal elicitors, the expression of this gene was found to be slightly upregulated in response to a bacterial pathogen. Response to elicitor was also reported in Nicotiana gene *NiLecRK1* (Sasabe et al. 2007). Upon studying the expression pattern of its Arabidopsis homologue At4G04960 it was found that the homologue gene also responded to bacterial pathogen *P. syringae*. However, homologue of the same gene in *Pisum sativum* (LecRLK, EU041719) has been demonstrated to be upregulated in salt stress (Joshi et al. 2010), while in silico expression data suggests downregulation of its Arabidopsis homologue in salt stress. Similar results have been found in case of Gossypium LecRLK (AY487461; Arabidopsis homologue: At3g53810) (Zuo et al. 2004), Nicotiana LecRLK (AB247455; Arabidopsis homologue: At3g53380) (Kanzaki et al. 2008) and Poplar LecRLK (AB030083; Arabidopsis homologue: At3g59700), where homologous genes have not been found to exhibit same results in the in silico analysis as they have been experimentally validated. In these cases, the differential gene expression and regulation could be the key controllers of the function of the gene instead of similar primary structure. Similar findings

have been reported in three isoforms of phospho enol pyruvate carboxylase (*PEPC*) gene where the isoforms differed in their expression pattern due to differential gene regulation (Lepiniec et al. 1993). In another case, Arabidopsis L type *LecRLK* gene, At1g70130, has been shown to be responsive to ABA, salt and osmotic stress (Deng et al. 2009) whereas the gene expression analysis revealed its slight upregulation only in salt stress while expression remained unchanged in case of ABA and osmotic stress which suggests the need of experimental validation of the in silico data available in the databases.

The analysis of the heat map generated for *LecRLK* group also revealed an interesting pattern. We observed a demarcation in expression pattern of *LecRLK* genes. The genes in this group seem to have developed differential and specific functions. We have observed that the genes that have no or low expression in different developmental stages of plant (for example, in case of rice L type *LecRLK*, Os07g03900, Os07g38250, Os07g18230, Os06g22290, Os02g48200, Os09g09510, Os02g19530, Os04g01890, Os08g03020, Os07g03920 and many more) have a role to play in environmental stress condition. Similar pattern was observed in other *LecRLK* groups as well. The behaviour is not a thumb rule as we found a few exceptions in which there was no expression of gene in any of the chosen conditions of stress, development or anatomy (Os04g34390, At1g11340, At1g61480, At1g61490) or few genes that expressed in stress conditions and developmental stages and/or anatomical features of plant (Os04g07200, Os01g57100, Os10g30530, Os01g11330, Os01g11280) but such instances were very few. Also, we have found that the genes which had higher expression in different developmental stages had high expression in different anatomical features as well. For example, in case of rice G type, Os06g35870, Os05g34390, Os06g41560, Os05g07080, Os06g41500, Os01g10700, Os04g07200 have high expression in different developmental stages. With exception of Os04g07200, all other genes were found to express well in different organs of plant as well.

In case of Arabidopsis G type, the overall expression of the genes overall is extremely low. Very few members of the gene family were moderately expressing in developmental stages of plant and anatomical features. In case of environmental conditions, biotic stress along with drought, heat and cold induce low level of expression, while hormones had little effect on expression of the gene family.

Conclusion

In the present study, we have analyzed *LecRLK* family. The study has increased our knowledge and understanding of fundamental aspects of the gene family. *LecRLK* family

was found to have large number of members amongst rice and arabidopsis. Although the number of this family of protein is huge, only a few members have been studied for their *in-planta* functions. Therefore, understanding of the functional relationship between members of this family is an important challenge. We have identified conserved motifs within the lectin domain of L and G type *LecRLK* families. We have laid more stress on expression analysis of the genes in different environmental conditions, different developmental stages and anatomical features of plant through publically available databases. The results presented in the current study emphasize the streamlined expression of individual gene of *LecRLK* family which otherwise has varied and diverse roles to play in plant's life. The results further encourage stress condition specific, tissue specific and developmental stage specific functional analysis of members of gene family and their further employment in various agricultural applications.

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